

PART A — QUESTIONS

AF22-001

Question:

Solve: $24x + 18 = 162$

- A) 5
- B) 6
- C) 7
- D) 8

AF22-002

Question:

If $22n - 13 = 141$, what is n ?

- A) 5
- B) 6
- C) 7
- D) 8

AF22-003

Question:

Simplify: $9(3x - 5) - 8x$

- A) $19x - 45$
- B) $27x - 45$
- C) $19x - 5$
- D) $35x - 45$

AF22-004

Question:

A company charges \$48 per delivery plus a \$25 account fee. Which expression represents the total charge for d deliveries?

- A) $48d + 25$
- B) $25d + 48$
- C) $73d$
- D) $48 - 25d$

AF22-005

Question:

If $f(x) = 19x - 58$, what is $f(8)$?

- A) 84
- B) 94
- C) 152
- D) 210

AF22-006

Question:

Which ordered pair satisfies the equation $y = 17x - 6$?

- A) (1, 12)
- B) (2, 28)
- C) (3, 45)
- D) (4, 60)

AF22-007

Question:

Solve: $26(x - 6) = 182$

A) 11

B) 12

C) 13

D) 14

AF22-008

Question:

A pattern begins: 14, 35, 56, 77, ... What is the next number?

A) 96

B) 97

C) 98

D) 99

AF22-009

Question:

Solve: $x/50 = 3$

A) 53

B) 100

C) 150

D) 200

AF22-010

Question:

A line has a slope of -19 and crosses the y-axis at 16. Which equation represents the line?

A) $y = 16x - 19$

B) $y = -19x + 16$

C) $y = 19x + 16$

D) $y = -16x + 19$

AF22-011

Question:

If 8 containers hold 312 parts, how many parts are in 5 containers at the same rate?

A) 185

B) 190

C) 195

D) 200

AF22-012

Question:

Solve: $390 - 36x = 246$

A) 3

B) 4

C) 5

D) 6

AF22-013

Question:

Which expression is equivalent to $120x - 55x + 14$?

- A) $65x + 14$
- B) $175x + 14$
- C) $65x - 14$
- D) $120x - 41$

AF22-014

Question:

A warehouse starts with 2,100 cartons. Each truck removes 150 cartons. Which equation gives the number of cartons C left after t trucks?

- A) $C = 2,100t - 150$
- B) $C = 150t + 2,100$
- C) $C = 2,100 - 150t$
- D) $C = 150t - 2,100$

AF22-015

Question:

If $y = -25x + 400$, what is y when $x = 14$?

- A) 40
- B) 50
- C) 350
- D) 750

AF22-016

Question:

Solve: $38x - 84 = 26x + 60$

- A) 10
- B) 11
- C) 12
- D) 13

AF22-017

Question:

Which value of x makes $22x + 9$ less than 273?

- A) 11
- B) 12
- C) 13
- D) 14

AF22-018

Question:

A table shows a linear relationship.

x : 0, 1, 2, 3

y : 9, 31, 53, 75

What is the rule?

- A) $y = 9x + 22$
- B) $y = 22x + 9$
- C) $y = 31x - 22$
- D) $y = x + 9$

AF22-019

Question:

Solve: $22(x + 4) = 19x + 127$

- A) 11
- B) 12
- C) 13
- D) 14

AF22-020

Question:

If $a = -18$ and $b = 15$, what is $3a + 4b$?

- A) -114
- B) -6
- C) 6
- D) 114

AF22-021

Question:

Which point is located on the y-axis?

- A) (17, 0)
- B) (0, -17)
- C) (-17, 5)
- D) (17, -5)

AF22-022

Question:

A worker sorts 288 items in 12 minutes. At the same rate, how many items can the worker sort in 7 minutes?

- A) 144
- B) 156
- C) 168
- D) 180

AF22-023

Question:

Solve for p : $F = 40p - 80$

- A) $p = F + 80$
- B) $p = (F + 80)/40$
- C) $p = 40(F + 80)$
- D) $p = F/40 - 80$

AF22-024

Question:

What is the slope of a line that goes through (10, 20) and (16, 110)?

- A) 13
- B) 14
- C) 15
- D) 90

AF22-025

Question:

Simplify: $92x - 31 - 81x + 47$

- A) $11x + 16$
- B) $173x + 16$
- C) $11x - 78$
- D) $173x - 78$

AF22-026

Question:

If $g(x) = x^2 + 8x$, what is $g(13)$?

- A) 169
- B) 208
- C) 273
- D) 338

AF22-027

Question:

A sequence follows the rule “multiply by 4, then subtract 15.” If the first term is 13, what is the second term?

- A) 37
- B) 45
- C) 52
- D) 67

AF22-028

Question:

Solve: $26(x - 5) + 34 = 242$

- A) 11
- B) 12
- C) 13
- D) 14

AF22-029

Question:

A line crosses the y-axis at -25 and rises 12 units for every 1 unit to the right. Which equation represents the line?

- A) $y = -25x + 12$
- B) $y = 12x - 25$
- C) $y = -12x - 25$
- D) $y = 25x + 12$

AF22-030

Question:

If $\frac{32}{48} = \frac{x}{144}$, what is x ?

- A) 84
- B) 90
- C) 96
- D) 108

AF22-031

Question:

Solve: $-45x + 225 = -360$

- A) -13
- B) -9
- C) 9
- D) 13

AF22-032

Question:

A stockroom starts with 7,500 labels. Each order uses 375 labels. Which expression gives the number of labels left after o orders?

- A) $7,500 + 375o$
- B) $7,500 - 375o$
- C) $375o - 7,500$
- D) $7,500 \div 375o$

AF22-033

Question:

Which equation has $x = 30$ as its solution?

- A) $2x + 19 = 77$
- B) $5x - 45 = 105$
- C) $x/15 + 8 = 11$
- D) $3x - 22 = 70$

PART B — ANSWER KEY + EXPLANATIONS

AF22-001 — Correct: B — Explanation:

Solve $24x + 18 = 162$. Subtract 18 from both sides: $24x = 144$. Divide by 24: $x = 6$. The most common mistake is subtracting 18 correctly but forgetting to divide by 24.

AF22-002 — Correct: C — Explanation:

Start with $22n - 13 = 141$. Add 13 to both sides: $22n = 154$. Divide by 22: $n = 7$. A common mistake is subtracting 13 instead of adding it.

AF22-003 — Correct: A — Explanation:

Distribute first: $9(3x - 5) = 27x - 45$. Then subtract $8x$: $27x - 8x - 45 = 19x - 45$. A common mistake is forgetting to multiply -5 by 9.

AF22-004 — Correct: A — Explanation:

The company charges \$48 for each delivery, so that part is $48d$. The \$25 account fee is added one time. The total charge is $48d + 25$. A common mistake is treating the account fee as the per-delivery charge.

AF22-005 — Correct: B — Explanation:

Substitute 8 for x : $f(8) = 19(8) - 58 = 152 - 58 = 94$. A common mistake is adding 58 instead of subtracting it.

AF22-006 — Correct: C — Explanation:

Test the choices in $y = 17x - 6$. For $(3, 45)$, $17(3) - 6 = 51 - 6 = 45$, which matches the y -value. A common mistake is choosing a point that is close but does not make the equation true.

AF22-007 — Correct: C — Explanation:

Solve $26(x - 6) = 182$. Divide both sides by 26: $x - 6 = 7$. Add 6: $x = 13$. A common mistake is adding 6 before undoing the multiplication by 26.

AF22-008 — Correct: C — Explanation:

The pattern increases by 21 each time: 14, 35, 56, 77. Add 21 to 77: 98. A common mistake is assuming the increase is 20 because the numbers are close to round numbers.

AF22-009 — Correct: C — Explanation:

$x/50 = 3$ means x divided by 50 equals 3. Multiply both sides by 50: $x = 150$. A common mistake is adding 50 and 3 instead of multiplying.

AF22-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is -19 and the y-intercept is 16, so the equation is $y = -19x + 16$. A common mistake is switching the slope and y-intercept.

AF22-011 — Correct: C — Explanation:

Find the unit rate: $312 \text{ parts} \div 8 \text{ containers} = 39 \text{ parts per container}$. For 5 containers: $39 \times 5 = 195$. A common mistake is multiplying 312 by 5 without finding the rate first.

AF22-012 — Correct: B — Explanation:

Solve $390 - 36x = 246$. Subtract 390 from both sides: $-36x = -144$. Divide by -36: $x = 4$. A common mistake is losing the negative sign after subtracting 390.

AF22-013 — Correct: A — Explanation:

Combine like terms: $120x - 55x = 65x$. The constant 14 stays the same. The expression becomes $65x + 14$. A common mistake is combining the constant with the variable terms.

AF22-014 — Correct: C — Explanation:

The warehouse starts with 2,100 cartons. Each truck removes 150 cartons, so t trucks remove $150t$ cartons. The number left is $C = 2,100 - 150t$. A common mistake is adding $150t$ even though cartons are being removed.

AF22-015 — Correct: B — Explanation:

Substitute $x = 14$ into $y = -25x + 400$: $y = -25(14) + 400 = -350 + 400 = 50$. A common mistake is treating $-25(14)$ as positive 350.

AF22-016 — Correct: C — Explanation:

Solve $38x - 84 = 26x + 60$. Subtract $26x$: $12x - 84 = 60$. Add 84: $12x = 144$. Divide by 12: $x = 12$. A common mistake is adding 84 to only one side.

AF22-017 — Correct: A — Explanation:

Solve $22x + 9 < 273$. Subtract 9: $22x < 264$. Divide by 22: $x < 12$. The only listed value less than 12 is 11. A common mistake is choosing 12, but $22(12) + 9 = 273$, not less than 273.

AF22-018 — Correct: B — Explanation:

The y-values increase by 22 each time x increases by 1, so the slope is 22. When $x = 0$, $y = 9$, so the y-intercept is 9. The rule is $y = 22x + 9$. A common mistake is using 9 as the slope because it is the first y-value.

AF22-019 — Correct: C — Explanation:

Distribute: $22x + 88 = 19x + 127$. Subtract $19x$: $3x + 88 = 127$. Subtract 88: $3x = 39$. Divide by 3: $x = 13$. A common mistake is forgetting to multiply 4 by 22.

AF22-020 — Correct: C — Explanation:

Substitute $a = -18$ and $b = 15$: $3a + 4b = 3(-18) + 4(15) = -54 + 60 = 6$. A common mistake is treating $3(-18)$ as positive 54.

AF22-021 — Correct: B — Explanation:

A point on the y-axis has $x = 0$. The point $(0, -17)$ is on the y-axis. A common mistake is choosing $(17, 0)$, which is on the x-axis.

AF22-022 — Correct: C — Explanation:

288 items in 12 minutes means $288 \div 12 = 24$ items per minute. In 7 minutes: $24 \times 7 = 168$. A common mistake is multiplying 288 by 7 without finding the rate first.

AF22-023 — Correct: B — Explanation:

Start with $F = 40p - 80$. Add 80 to both sides: $F + 80 = 40p$. Divide by 40: $p = (F + 80)/40$. A common mistake is dividing F by 40 first and then adding 80.

AF22-024 — Correct: C — Explanation:

Slope equals change in y divided by change in x . From $(10, 20)$ to $(16, 110)$, y increases by 90 and x increases by 6. The slope is $90 \div 6 = 15$. A common mistake is using 90 as the slope without dividing by the change in x .

AF22-025 — Correct: A — Explanation:

Combine like terms: $92x - 81x = 11x$, and $-31 + 47 = 16$. The simplified expression is $11x + 16$. A common mistake is writing $11x - 16$ because the positive 47 is missed.

AF22-026 — Correct: C — Explanation:

Substitute 13 for x : $g(13) = 13^2 + 8(13) = 169 + 104 = 273$. A common mistake is calculating 13^2 as 26 instead of 169.

AF22-027 — Correct: A — Explanation:

Start with 13. Multiply by 4: 52. Then subtract 15: 37. A common mistake is subtracting 15 first and then multiplying.

AF22-028 — Correct: C — Explanation:

Solve $26(x - 5) + 34 = 242$. Distribute: $26x - 130 + 34 = 242$. Combine: $26x - 96 = 242$. Add 96: $26x = 338$. Divide by 26: $x = 13$. A common mistake is forgetting to combine -130 and $+34$.

AF22-029 — Correct: B — Explanation:

The line rises 12 units for every 1 unit to the right, so the slope is 12. It crosses the y-axis at -25 , so the equation is $y = 12x - 25$. A common mistake is switching the slope and intercept.

AF22-030 — Correct: C — Explanation:

Use the proportion $32/48 = x/144$. Since $48 \times 3 = 144$, multiply 32 by 3: $x = 96$. A common mistake is adding 96 to the numerator because 48 becomes 144.

AF22-031 — Correct: D — Explanation:

Solve $-45x + 225 = -360$. Subtract 225 from both sides: $-45x = -585$. Divide by -45: $x = 13$. A common mistake is giving -13 because both sides contain negative numbers.

AF22-032 — Correct: B — Explanation:

The stockroom starts with 7,500 labels. Each order uses 375 labels, so o orders use $375o$ labels. The number left is $7,500 - 375o$. A common mistake is adding $375o$ even though labels are being used.

AF22-033 — Correct: B — Explanation:

Test $x = 30$. In choice B, $5(30) - 45 = 150 - 45 = 105$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.